

Capstone Program Brief — Shark Tank Edition

Build a working, evaluated AI system on a real enterprise problem. Then pitch it to a panel of industry practitioners who will grade the engineering *and* the business case.

15

Capstone projects

5

Enterprise domains

7

Capability areas

30_{min}

Pitch: 20 + 10 Q&A

HOW TO READ A PROJECT

Seven capability areas, one integrated build

Every project stitches together several of the areas below. Each project card lists the specific ones it draws on. A capstone must integrate a **minimum of four**.

M1 LLM Engineering — APIs, structured output, routing, cost & latency

M2 RAG Systems — ingestion, hybrid search, re-ranking, RAGAS

M3 Fine-Tuning & Alignment — SFT, LoRA/QLoRA, DPO, embeddings

M4 AI Agent Engineering — LangGraph, tools, memory, multi-agent

M5 Multimodal AI — document AI, vision-language, audio

M6 LLMOps — serving, CI/CD, observability, caching

M7 Evaluation & Safety — LLM-as-Judge, red-team, guardrails, bias

● Insurance

● Banking

● Healthcare

● Supply Chain

● Logistics

Intelligent Motor Claims Processing Platform

P01

BUSINESS PROBLEM

A motor insurer handles 800 claims/day at a 10-day settlement TAT with a 12% suspected-fraud rate. The manual pipeline is slow and expensive.

WHAT TO BUILD

End-to-end pipeline: multimodal ingestion (damage photos + repair estimates + medical bills) → structured extraction → fraud ML model → policy-coverage verification via RAG → agentic settlement recommendation → LLM-generated customer letter. Full audit trail, human-in-the-loop for disputed cases.

M2

M4

M5

M6

M7

Grades on: functional pipeline on real-format test claims · fraud AUC & precision-at-threshold · RAG faithfulness > 0.85 · audit + HITL correctness · projected TAT reduction and cost savings.

AI-Powered Underwriting Assistant

P02

BUSINESS PROBLEM

Underwriters spend 2–3 hours per commercial policy application. A team of 10 clears 50 applications/day; the target with AI assistance is 150/day.

WHAT TO BUILD

Multimodal pipeline: ingest application form + financial statements + property photos → extract structured applicant and risk data → enrich with external signals (mock APIs: loss ratios, location risk) → RAG over underwriting guidelines → generate a structured Underwriting Report (risk score, key factors with evidence, coverage recommendation, premium range, exclusions). Bias audit on scoring, HITL for borderline cases.

M1

M2

M4

M5

M7

Grades on: field-level extraction precision · correct risk rules cited per scenario · report quality (LLM-as-Judge) · demographic-proxy bias audit · < 5 min end-to-end latency.

Policy Servicing & FNOL Assistant

P03

BUSINESS PROBLEM

The service desk is flooded with policy-wording questions, endorsement requests, and first-notice-of-loss intake — slow, inconsistent, and costly to staff.

WHAT TO BUILD

Customer-facing assistant: RAG over policy wordings and benefits, agentic tools for transactional queries (premium due, coverage check, endorsement eligibility via mock APIs), structured FNOL intake with ticketing, and escalation to a human with a full context summary. Guardrails (no financial advice, no policy modification), 20+ adversarial red-team prompts, hallucination measurement. LLMOps deployment.

M2

M4

M6

M7

Grades on: RAG faithfulness on policy Q&A · tool-calling correctness on transactional queries · red-team + guardrail effectiveness · deployment with observability.

Agentic Loan Origination System

P04

BUSINESS PROBLEM

Retail loan processing takes 5 business days across 4 teams. Straight-through processing for low-risk profiles is not yet automated.

WHAT TO BUILD

Multi-agent LangGraph system: DocumentCollection → KYCVerification (multimodal ID extraction + sanctions check) → CreditAssessment (bureau + income verification) → RegulatoryCompliance (RAG over RBI lending guidelines) → Decision (approve / reject / escalate with explainable rationale). Full checkpointing, HITL for borderline cases, complete audit log for regulatory review.

M1

M2

M4

M5

M6

Grades on: all 5 agents functional with correct handoffs · KYC field extraction precision · correct regulatory citations · complete audit log · Dockerised with a working API endpoint.

Real-Time Fraud Explanation Engine

P05

BUSINESS PROBLEM

A payment network's model flags transactions but explains nothing. Investigators spend 8 minutes per flag gathering context. Explanations must generate in under 10 seconds.

WHAT TO BUILD

Classical ML fraud model (XGBoost on transaction data) plus an LLM explanation layer: on a flag, an agent gathers context (merchant history, spending pattern, device signals), writes a human-readable investigation brief, and suggests the top 3 investigation actions. Full LLMOps: vLLM serving, LangSmith observability, latency-SLA monitoring.

M1

M4

M6

M7

Grades on: fraud AUC & precision at the business-optimal threshold · explanation quality (LLM-as-Judge: accuracy, actionability, clarity) · end-to-end P95 < 10s · observability dashboard · hallucination rate < 5%.

BUSINESS PROBLEM

Compliance officers spend 6+ hours weekly searching regulatory documents. One wrong interpretation carries real penalty exposure.

WHAT TO BUILD

RAG over 60+ RBI, SEBI, and IRDAI documents with hybrid search and a fine-tuned embedding model (trained on BFSI regulatory text). Features: citation-grounded Q&A, a regulatory change tracker (what changed between two versions of a circular), and a compliance checklist generator (given a business activity, produce the applicable checklist). Full RAGAS evaluation.

M2

M3

M6

M7

Grades on: RAGAS faithfulness > 0.88 & answer relevance > 0.82 · retrieval lift of the fine-tuned embedding vs generic · change-tracker accuracy · hallucination < 3% · UI/API usability.

Clinical Prior-Authorization Automation Platform

P07

BUSINESS PROBLEM

Prior authorization for procedures and medications takes days of manual review of clinical notes against payer medical-necessity policy. Wrong denials trigger costly appeals.

WHAT TO BUILD

Multimodal ingestion (referral forms, clinical notes, lab PDFs) → structured clinical-data extraction → RAG over payer medical-necessity policies → agentic decision (approve / pend / deny with cited criteria) → auto-generated determination letter. HITL for denials, full audit trail. Safety: no clinical advice, strict PHI-handling guardrails. Use only synthetic records.

M2

M4

M5

M6

M7

Grades on: extraction precision on clinical docs · correct policy criteria cited per case · decision rationale quality (LLM-as-Judge) · audit + HITL correctness · guardrail effectiveness & hallucination rate.

Medical Coding & Claims Adjudication Assistant

P08

BUSINESS PROBLEM

Coders assign ICD-10 / CPT codes from charts by hand. Errors drive claim denials and revenue leakage.

WHAT TO BUILD

Ingest clinical documentation (multimodal) → suggest ICD-10 / CPT codes with supporting evidence spans → RAG over coding guidelines and payer rules → adjudication check (flag mismatches, missing modifiers, unbundling) → structured coding worksheet. Fine-tune an embedding model on medical terminology. Error / bias audit on suggestions.

M2

M3

M4

M5

M7

Grades on: code-suggestion precision/recall vs a gold set · evidence grounding · correct rule citations · adjudication-flag accuracy · hallucination rate.

Patient Discharge & Care-Navigation Assistant

P09

BUSINESS PROBLEM

Discharge summaries are dense and clinical. Patients miss follow-ups and warning signs, driving avoidable readmissions.

WHAT TO BUILD

Ingest discharge notes → generate a plain-language, structured discharge summary (medications, follow-ups, warning signs) → an agentic care-navigation assistant answers patient questions via RAG over condition and medication information → escalation to the care team. Safety-critical guardrails (no diagnosis, no dosage changes), red-teaming, and readability + accuracy evaluation. LLMOps deployment.

M1

M2

M4

M6

M7

Grades on: summary faithfulness to source · readability score · escalation/tool correctness · red-team + guardrail effectiveness · hallucination rate.

Supplier Risk & Contract Intelligence Platform

P10

BUSINESS PROBLEM

Procurement teams manually review supplier contracts, financials, and news for risk. It's slow and red flags slip through.

WHAT TO BUILD

Multimodal ingestion (contract PDFs, financial statements, certifications) → structured clause and obligation extraction → RAG over policy and compliance standards → agentic risk scoring across financial, compliance, concentration, and ESG signals (mock APIs) → a risk brief with cited evidence. Consistency / bias audit on scoring, HITL for high-risk suppliers.

M2

M4

M5

M7

Grades on: clause-extraction precision · correct standard citations · risk-brief quality (LLM-as-Judge) · scoring consistency & bias audit · end-to-end latency.

Demand-Planning Copilot (Multi-Agent Forecasting)

P11

BUSINESS PROBLEM

Planners reconcile forecasts across sales, promotions, seasonality, and external signals by hand — siloed, slow, and hard to explain.

WHAT TO BUILD

Multi-agent system: DataAgent (pull mock sales/inventory) → ForecastAgent (classical model + LLM adjustment for known events) → AnomalyAgent (flag demand shifts) → NarrativeAgent (explain the forecast and recommend actions). LangGraph with memory and checkpointing. Observability and output evaluation.

M1

M4

M6

M7

Grades on: agent workflow completeness & handoffs · forecast-explanation quality · anomaly-flag accuracy · observability dashboard · hallucination on generated narratives.

Procurement Invoice & PO Reconciliation Engine

P12

BUSINESS PROBLEM

Three-way matching (PO, goods receipt, invoice) is manual, causing exceptions, duplicate payments, and invoice fraud.

WHAT TO BUILD

Multimodal ingestion of invoices, POs, and receipts → structured line-item extraction → agentic three-way match (flag mismatches, missing items, price variance) → RAG over procurement policy for exception rules → structured exception report with suggested resolution. Duplicate / fraud detection, audit trail, HITL. Dockerised endpoint.

M2

M4

M5

M6

M7

Grades on: line-item extraction precision · match accuracy · correct policy-rule citations · exception & duplicate-flag correctness · audit + deployment.

Shipment Exception Management Agent

P13

BUSINESS PROBLEM

Shipments hit exceptions — delays, damage, customs holds, misroutes. Ops teams triage across multiple systems by hand, so resolution is slow.

WHAT TO BUILD

Agentic system: ingest exception events + tracking/EDI data → classify the exception → RAG over SOPs and carrier rules for the right resolution → agentic action (mock APIs: reroute, notify, reschedule, file claim) → customer comms and escalation. Memory, loop detection, and retry/error handling. LLMOps with observability.

M1 M2 M4 M6 M7

Grades on: exception-classification accuracy · correct SOP citations · tool-calling correctness · escalation safety · latency + observability.

Freight Document & Customs Compliance Platform

P14

BUSINESS PROBLEM

Cross-border shipments need a bill of lading, commercial invoice, packing list, and correct HS classification. Manual handling is error-prone and causes customs delays.

WHAT TO BUILD

Multimodal ingestion of freight documents → structured extraction and cross-document consistency checks → RAG over customs/trade regulations and HS-code rules → HS-code suggestion with cited basis → compliance checklist and discrepancy report. Optional fine-tuned embedding on trade terminology. HITL and audit trail.

M2 M3 M4 M5 M7

Grades on: extraction precision · cross-document consistency detection · correct HS/regulation citations · checklist completeness · hallucination rate.

BUSINESS PROBLEM

Dispatch and support handle a flood of "where is my order", failed deliveries, address corrections, and reschedules at scale.

WHAT TO BUILD

Customer + ops assistant: RAG over delivery policies and FAQs, agentic tools (track shipment, reschedule, update address, initiate return via mock APIs), failed-delivery intake with ticketing, and escalation with full context. Guardrails, red-teaming, and bias audit. CI/CD deployment, semantic caching, and observability.

M2

M4

M6

M7

Grades on: RAG accuracy · tool-calling correctness on transactional queries · red-team + guardrail effectiveness · LLMOps (CI/CD, caching, observability) · demo & business-case quality.

GROUND RULES

Team & execution guidelines

Read these before you pick a project. The panel assumes you have followed them.

Team & scope

- **Max 3 per team.** Every member owns at least one capability area and must defend it in Q&A.
- **One project per team.** Pick from the 15 above — no custom scope without facilitator sign-off.
- **Integrate ≥ 4 of the 7** capability areas. The card's listed areas are the floor, not a ceiling.
- **Synthetic data only.** No real PII or PHI. Generate realistic, real-format test data.

What "done" means

- **It runs.** A working end-to-end pipeline or agent — not slideware or a notebook of fragments.
- **It's measured.** Report real numbers against the card's grading criteria — accuracy, latency, cost.
- **It's safe.** At least one evaluation/safety component: guardrails, red-team, hallucination or bias audit.
- **It's reproducible.** Code repo + README that lets the panel run it.

Required deliverables — submit before the pitch

- Code repository with a README (setup, run instructions, architecture diagram).
- Evaluation report with the metrics named in your project's grading criteria.
- A working demo — live or a recorded fallback in case of network failure.
- The 20-minute Shark Tank deck (PDF).

THE PITCH

Shark Tank presentation format

30 minutes total per team: a 20-minute presentation followed by 10 minutes of Q&A from a panel of industry practitioners who grade both the engineering and the business viability.

20-minute presentation — suggested split

3 min

00:00–03:00

Problem & business case.

The pain, who feels it, and the cost of the status quo.

4 min

03:00–07:00

Solution architecture. The pipeline, the agents, and which capability areas you integrated and why.

7 min

07:00–14:00

Live demo. Show it working on real-format inputs. This is the centrepiece — spend the time here.

4 min

14:00–18:00

Evaluation & results. The numbers: accuracy, latency, cost, hallucination/bias findings.

2 min

18:00–20:00

Safety story & the ask.

Guardrails and limits, then quantified business impact and what you'd build next.

10-minute Q&A — what the panel will probe

- **Does it actually work?** Expect a request to try an edge case or an input you didn't rehearse.
- **Are the numbers real?** How you measured, on what test set, and where it fails.
- **Would a business buy it?** Cost per transaction, TAT/headcount impact, and the path to production.
- **Is it safe?** How you catch hallucinations, block misuse, and keep a human in the loop.
- **Everyone answers.** The panel may direct questions to specific members on the areas they own.

Bring a recorded demo as a fallback. A failed live demo with no backup costs you the biggest block of marks.

HOW YOU'RE SCORED

Evaluation rubric

Each criterion is scored 1–5 by the panel, then weighted. Total out of 100.

Criterion	Weight	What "excellent" (5) looks like
Functionality	25%	A complete pipeline runs end-to-end on real-format inputs, including on an input the panel supplies live. No dead ends.
Technical depth & integration	20%	Four or more capability areas integrated cleanly, with sound engineering choices the team can justify under questioning.
Evaluation rigor & safety	15%	Metrics measured properly against a real test set; guardrails, red-team, and hallucination/bias findings reported honestly, including failure modes.
Usefulness & business impact	15%	Impact is quantified — TAT, cost, headcount, or risk reduction — with a credible path to production.
Novelty & innovation	15%	A non-obvious approach, a smart architecture choice, or a capability that goes meaningfully beyond the brief.
Presentation & Q&A	10%	Clear, well-paced pitch; a demo that lands; confident, specific answers from every team member.

Scoring: 5 excellent · 4 strong · 3 competent · 2 partial · 1 minimal. Weighted score = $\sum (\text{criterion score} \div 5 \times \text{weight})$.

Pre-pitch submission checklist

- ☐ Code repo + README submitted and runnable
- ☐ Evaluation report with the card's named metrics
- ☐ Synthetic data only — no real PII/PHI
- ☐ Every member can defend their area in Q&A
- ☐ ≥ 4 capability areas integrated
- ☐ At least one safety/eval component included
- ☐ 20-min deck (PDF) + recorded demo fallback
- ☐ Business impact quantified in the deck